MILITARY STANDARD

ASSEMBLY PROCEDURES, GIMBAL



FSC 1420

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Assembly Procedures, Gimbal

MIL-STD-2081(AS)

1. This Military Standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this standard should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

This standard has been prepared to define inspection and assembly procedures for Sidewinder Gimbal Assemblies and is suitable for citation in request for proposals, invitation for bids, contracts and purchase orders, or governing specifications.

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1. SCOPE

1.1 AI Purpose. This standard describes inspection and assembly procedures for Sidewinder gimbal assemblies, Figures 1 and 2. The gimbal assembly provides angular freedom for the Sidewinder missile gyro and is an integral subassembly unit of the Guidance and Control Section, WGU-4/B and WGU-4A/B.

1.2 AI Scope. Five sections comprise this standard. The sections are as follows:

a. Scope.

b. Referenced documents.

c. Definitions.

d. General requirements.

e. Detailed requirements.

1.3 AI Paragraph coding. Each paragraph and table is coded as follows:

M meaning Mandatory; i.e., use instruction as stated, except for tooling.

MXM meaning Mandatory Except for Method.

OP meaning Optional Process.

AI meaning Advisory Information.

2. REFERENCED DOCUMENTS

2.1 MXM Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

Federal

O-C-265 - Chemicals, Analytical; General Specification for.

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O-E-760	-	Ethyl Alcohol (Ethanol); Denatured Alcohol: Proprietary Solvents and Special Industrial Solvents.

Military

MIL-L-6085	-	Lubricating Oil, Instrument, Aircraft, Low
MIL-C-81302	-	Cleaning Compound, Solvent, Trichlorotrifluoro-ethane.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

639AS537	-	Post, Gimbal.
639AS551	-	Ring, Gimbal.
639AS624	-	Gimbal, Outer.
639AS627	-	Shim, Bearing, Gimbal.
639AS750	-	Screw, Gimbal.
639AS1469	-	Screw, Shipping.
639AS1471	-	Retainer, Shipping.
639AS1479	-	Nut, Shipping.
639AS1487	-	Adhesive, Epoxy.
639AS3143	-	Bearing, Spin.
639AS3168	-	Bearing, Gimbal.

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639AS3169 - Gimbal Assembly.

639AS3172 - Gimbal, Inner.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-874 - Clean Room Conditions.

AD-883 - Gyro Specialist and Gyro Worker Qualification.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 MXM Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AFBMA Standards - Annular Bearings Engineers Committee of the Anti-Friction Bearing Manufacturers Association (AFBMA) (Twelve Sections).

(Applications for copies of AFBMA Standards may be addressed to the Anti-Friction Bearing Manufacturers Association, 60 East 42nd, Rm 2825, New York, New York 10017.)

3. DEFINITIONS (Not Applicable)

4. GENERAL STATEMENT OF REQUIREMENTS

4.1 M End product. The end product gimbal assembly shall comply with all listed applicable documents as set forth in Section 2 of this standard and as specified herein.

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4.1.1 M Production techniques. Production techniques are to be controlled by process sheets. This is true for all machining, inspection, and assembly phases. The instructions herein are intended to ensure acceptable quality products.

4.1.2 M Process documents. All manufacturing, inspection, testing, and assembly techniques and procedures are to be documented and available to the particular work station involved. These procedures are to conform to the basic principles of inspection and assembly as set forth in this standard.

4.1.3 M Clean room. Clean room assembly conditions are to be in accordance with AD-874. Assembly personnel are to be qualified in accordance with AD-883.

4.2 MXM Equipment. This section lists the commercial products and equipment used during various manufacturing stages of Gimbal Assembly, Drawing 639AS3169. The models or products in effect at date of issue of this standard are considered to be acceptable for use in manufacturing this assembly. However, any changes in the form of modifications may render the equipment unacceptable for use. Authorization to use the modified equipment may be obtained from the cognizant Government representative. Divided into two categories, the equipment and products are as follows:

4.2.1 AI Mechanical equipment.

4.2.1.1 AI (1) Federal Electro-probe Electronic Test Indicator, Model-230P-3-R2, Federal Products Corporation, Providence, Rhode Island 02901. Vendor Code Ident 21938. (2) Cleveland Indi-Ac, Model JR-10BAC, Bendix Corporation. The Cleveland Instrument Plant of Industrial Metrology, Cleveland, Ohio 44131. Vendor Code Ident 99444.

4.2.1.2 AI American Optical Microscope, American Optical Instrument Division, Buffalo, New York 14215. Vendor Code Ident 78039. Variable power zoom models 10 power (10X) to 20X or higher are suggested.

4.2.1.3 AI Deltronic Tenth-Plug Gage Set, Hi-Precision Grinding, Incorporated, Costa Mesa, California 92626. Vendor Code Ident 96674.

4.2.1.4 AI Demagnetizer: 4 to 8 inch pass-through diameter toroidal coil, operating on 60 hertz (Hz), 115 volts alternating current and 8 to 12 amperes. R. B. Annis Company, 1101 North Delaware Street, Indianapolis, Indiana 46202. Vendor Code Ident 03629.

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4.2.2 AI Auxiliary items.

4.2.2.1 AI Cobehn Spray, Cobehn, Incorporated, 226 Passaic Avenue, Fairfield, New Jersey 07006. Vendor Code Ident 04882.

4.2.2.2 AI Acetone in accordance with O-C-265, Reagent Grade.

4.2.2.3 AI Ethanol, Grade I in accordance with O-E-760, Class A or B.

4.2.2.4 AI Trichlorotrifluoroethane, in accordance with MIL-C-81302, Type I. All references to Freon within this specification are considered interchangeable with trichlorotrifluoroethane.

4.2.2.5 AI Plastic Stem Cotton Swabs.

4.2.2.6 AI Hamilton Microliter Syringe, No. 710 with Cheney Adaption (Hypodermic syringe with barrel graduated in microliters; No. 26 needle), Hamilton Company, P.O. Box 307, Whittier, California 90608. Vendor Code Ident 22582.

4.2.2.7 AI Texwipe, Foamwipe, Foam-Swabs, Texwipe Company, Hillsdale, New New Jersey 07642. Vendor Code Ident 21994.

4.2.2.8 AI Adhesive, Epoxy, Drawing 639AS1487.

4.2.2.9 AI Syringe 5 cubic centimeters (cc) with Luerlock Adapter, Filter Holder, Millipore Filter Company, Bedford, Massachusetts 01730. Vendor Code Ident 08071.

4.2.2.10 AI Towels, Nylon, Lintless.

4.3 AI Solvent warning. Post the following warning notice at each work station:

WARNING AI

Prolonged breathing of solvent vapors can be dangerous. Proper ventilation is necessary in any area where it is used. Acetone and ethanol are highly flammable liquids. Fire-hazard precautions and proper ventilation are recommended in any area where they are used. Treat all solvents with caution since prolonged breathing of vapors or skin contact is injurious.

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4.3.1 M Solvent purity. Use only high-purity, reagent-grade solvents in all processes specified herein.

4.4 AI Preassembly information.

4.4.1 AI Shop accessory (SA) tooling. This paragraph provides information on SA drawings of recommended SA equipment used in various phases of assembly. Substitutes or innovations may be used if results prove satisfactory. Familiarity with SA drawings is useful and each operator should be acquainted with the SA tool to the extent that they know when the tool is damaged or not capable of performing the operation correctly.

4.4.1.1 AI Naval Air Systems Command (NAVAIR) SA fixtures. SAs bearing a six-digit basic number are various forms of special tools controlled by the Naval Weapons Center, China Lake, California. The following is a listing of NWC/NAVAIR SA drawing numbers:

SA 492478-3	-	Spanner, Tooling Gyro.
SA 493213	-	Inspection Fixture, Gimbal Axis Alignment (Gimbal to Spin Axis).
SA 493214	-	Inspection Fixture, Gimbal Axis Alignment (Gimbal to Gimbal).
SA 493245	-	Spin Bearing Preload Fixture.
SA 493289	-	Shop Accessories (Applied to Gimbal Support Post).
SA 495231	-	Wrench Kit, Gyro.
SA 495239	-	Gimbal Axis Torque Tester.
SA 495242	-	Gimbal Assembly Fixture.

4.4.1.2 AI Vendor SA drawings. SAs bearing a seven-digit basic number are various forms of special tools designed by a vendor under Government contract and are as follows:

SA 2523897	-	Assembled Gimbal Bearing Preload Test Fixture Assembly and Details.

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4.4.2 M Summary on tooling. To minimize fretting of parts, hardened tools in contact with gimbal hardware are required for the gimbal assembly.

4.4.2.1 MXM SAs and tooling inspection. Inspection of SAs and associated tooling for contamination and damage at critical surfaces is required at the start of every 8 hour-work shift to prevent parts damage. This is the responsibility of personnel using the tooling. A 20X (minimum) microscope is used for this check, where feasible, since the unaided eye cannot adequately detect minute (yet damaging) contamination and tool degradation. Successful completion of gimbal assemblies is a direct function of high-level cleanliness and vigilant quality control.

4.4.2.2 M Cleanliness. Maximum cleanliness is required during all assembly periods (see 4.5.7).

4.4.3 M Component inspection.

4.4.3.1 M Contamination. Perform examination for loose foreign materials with a 10X (minimum) microscope. Perform necessary cleaning to remove foreign material before using hardware in assembly.

4.4.3.2 M Burrs, slivers, feather edges. Perform examination on prime datum surfaces for these defects with the aid of a 10X microscope. All remaining surfaces shall be examined for these defects with the aid of a 4X microscrope. Remove loose burrs, slivers, and feather edges to preclude later contamination within the bearings.

4.4.3.3 M Nicks, dents, and bulges on functional surfaces. Perform examination for these defects with the aid of a 10X microscope. Repair deformations such as nicks, dents, and bulges resulting in raised material on functional surfaces such that they will meet drawing requirements for surface quality and dimensional accuracy.

NOTE: AI The use of a variable power zoom microscope allows for easy finding of trouble spots at 10X magnification, and then makes a thorough examination of questionable areas at 20X magnification.

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4.4.4 M Overhaul and repair policy. Disassembly for repair, readjustment, overhaul, or replacement is to be accomplished under the same stringent cleanliness requirements as the original assembly with the added requirement that all disassembly debris be thoroughly and completely removed before any reassembly occurs.

4.5 AI Assembly data.

4.5.1 MXM Adhesive application. Because successful assembly depends on many detailed assembly procedures, the following methods of application are recommended. Strict adherence to the instructions will minimize poor adhesion and subsequent failure of a unit. Exercise discretion when applying adhesive so that no contamination occurs. Clean the surfaces prior to application of adhesive, and if contaminated, remove all oil, grease, residues, and dirt. Mix and cure the adhesive in accordance with manufacturer's instructions, unless otherwise specified herein, or on the applicable drawings. Heat cures are not to exceed 90 degrees Celsius (°C). Use of colorant is optional.

4.5.1.1 MXM Solvents. Cobehn, Freon TF, acetone, or ethanol applied with swabs as lint-free as possible are used for solvent cleaning. When using Cobehn solvent or any solvent on cemented and assembled parts, exercise manual control in a manner that will not flood the spray-cleaned surfaces; such flooding will destroy uncured adhesive bonding. Since no swab is completely lint-free, keep all swabs in clean containers until used. Upon use, discard the swabs in covered disposal containers. Use only small quantities of solvent at a time. When using, ensure that the solvents are kept in small clean containers that yield clean solvents at all times. Do not use the same solvents or swabs more than once since this will result in contamination (see 4.3.1). Solvents and air used for solvent spray must be filtered to remove particulate contamination greater than 0.5 micron.

4.5.1.2 MXM References. "Acetone swab", "ethanol swab", or "damp swab" means that a dry swab is dipped in the appropriate solvent in such a manner that, upon application, the swab is damp but not dripping. This is accomplished by dipping a clean swab in the solvent and removing the excess by blotting on a folded Foamwipe or equivalent. Since bonding in many areas will be adversely affected, avoid using dripping swabs during any assembly operation where adhesive damage may occur.

4.5.2 M Contamination prevention. Avoid unnecessary cleaning of spin bearings, gimbal bearings, and the finished gimbal assembly. Handle unmounted bearings under clean hoods when not in a protective container or in clean work stations as defined in AD-874. Particular attention is drawn to the spin bearings which

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contain phenolic cages that are impregnated with lubricating oil to act as an oil reservoir for long storage. The bearings are grease injected to provide a uniform film of lubricant over the working surfaces of the bearing. Cleaning with solvents may remove sufficient grease and lubricating oil as to cause shortened bearing life. Do not use a swab near enough to any bearing to cause contamination of the bearing unless otherwise specified herein. Contaminated spin bearings are to be relubricated by a qualified lubrication facility. Remove excess oil from gimbal bearings by centrifuging at 500 gravity units (g) for 2 minutes before assembly to reduce contamination during assembly, or by vacuum removal as specified in 5.1.2.22.2. Do not handle spin or gimbal bearings in any manner except with bearing handling tools or finger cots. Demagnetize all bearings prior to cleaning and use only nonmagnetic bearing handling tools. All tools and fixtures that are used to handle the bearings are to be clean.

4.5.2.1 M Bearing coding. In order to provide proper spin and gimbal bearing fits, it is necessary that the inside diameter (ID) and outside diameter (OD) of these parts be known to an accuracy compatible with the fit desired. To preserve the integrity of the bearing, it is necessary that the bearings be coded for size prior to delivery to assembly position. Keep handling of the bearing to a minimum.

4.5.3 AI Gimbal hardware coding. To reduce bearing handling damage, the coding of gimbal trunnions and bores is recommended. These should match the gimbal and spin bearing coding, except individual gimbal parts will not be categorized as a part possessing certain size, such as is done with the bearings. This will allow assembly to be accomplished in a manner making the greatest use of bearing size variation.

4.5.3.1 AI Gimbal hardware and bearing matching. Mate preselected groups of hardware and bearings for fit. It may be necessary in some cases to trade off bearings for bearings of the same code to achieve the proper fit (i.e., compensate for variations within the code). Measurements for mating fits are to be made at similar temperatures within a 5°C range.

4.5.3.2 AI Gimbal bearing lubrication. Immediately prior to assembly, lubricate the gimbal bearings as specified in 5.1.2.22 and test for bearing noise quality in accordance with Drawing 639AS3169.

4.5.4 M Contamination prevention. In all assembly procedures, constant vigilance and examination under 10X magnification is required to maintain contamination-free assemblies. Liberal use of vacuum cleaning probes and fine needle-like tweezers is required for removal of contamination as it occurs. If contamination is allowed to remain for later removal, the contamination can work into the system. This can cause catastrophic failure of the assembly.

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4.5.5 M Quality assurance. Quality assurance of the gimbal assembly is governed by the requirements of Drawing 639AS3169.

4.5.6 M Instructions. All contractor assembly and test procedures involving the gimbal assembly are governed by the instructions set forth in this standard and the applicable drawings. Contractor assembly and test procedures pertaining to mandatory parts of this standard are subject to review at any time by the cognizant Government representative to ensure compliance.

4.5.7 M Assembly operations. All assembly personnel concerned with a particular assembly operation of this standard should be thoroughly familiar with that process. Assembly personnel are to be trained to be constantly alert to maintain absolute cleanliness control.

4.5.8 M Document control. Maintain a copy of this standard immediately outside clean room areas for ready reference at all times. However, no standard is to be inside the clean room unless permanently sealed in plastic containers or printed on vellum or glossy plastic sheets by an acceptable method.

4.5.9 MXM Ancillary requirements. When warm air is referred to, it refers to air temperature of at least 65°C but not to exceed 120°C. Clean filtered air refers to air that meets the requirements of AD-874.

4.5.10 AI These instructions are based on experimentation and research as opposed to production methods.

NOTE: AI The procedures herein are based on the assembly of one unit at a time. Production procedures may dictate more conservative use of time without violating requirements for successful assembling. Since ease of production is the primary reason for the existence of this standard, some innovations may be necessary. Deviations from the basic principles are not allowed. Sequence of operations may be varied to suit production requirements provided that dimensional and functional integrity is not impaired.

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5. DETAILED STATEMENT OF REQUIREMENTS

5.1 AI Assembling gimbal assembly. This section describes the assembly procedures for Gimbal Assembly Drawing 639AS3169 (see Figure 1).

5.1.1 AI Assembling inner axis assembly (see Figure 3). Proceed as follows:

5.1.1.1 AI Inner gimbal. Place inner gimbal on preselected pin of Deltronic plug gage set. Use gage pin that provides a light wring-fit so that no axial misalignment can occur.

5.1.1.2 AI Placement and leveling. Place setup in V-block and level trunnions within 0.000025 inch. Rotate inner gimbal 180 degrees on the tenth-gage pin and relevel within 0.000025-inch total indicated readout (TIR) without disturbing indicator setting; note the difference. Verify that intersection is within drawing requirements. Use electroprobe test indicator with 2.5 gram maximum contact pressure.

5.1.1.3 AI Outer gimbal. Place outer gimbal (Item 2, Figure 3) on 3/16-inch Deltronic pin with light wring-fit and place setup on V-block and check intersection as specified in 5.1.1.2. Verify that the geometric axis of nominal 0.0937-inch trunnions is within drawing requirements.

5.1.1.4 MXM Gimbal assembly. Assemble outer gimbal (Item 2, Figure 3), two shims (Item 3, Figure 3), two gimbal screws (Item 5, Figure 3), and two gimbal bearings (Item 4, Figure 3) to inner gimbal (Item 1, Figure 3). Select gimbal bearings that provide 0.000000 to 0.000050-inch clearance between bearing OD and its mating bore, and 0.000000 to 0.000050-inch clearance between bearing bore and its mating shaft. Select two shims slightly thicker than normally required to preload outer gimbal to inner gimbal. Considering that mating diameters are not perfectly round, bearing-fit clearances refer to maximum OD and minimum ID differences. Torque gimbal screws 16-18 inch-ounces. Do not cement screws.

CAUTION MXM

During the assembly of the shims on the inner gimbal, mount shims with the ID chamfer facing the fillet radius of the trunnion shoulder.

5.1.1.5 M Axial play. Measure total axial play using Fixture SA 2523897. Adjust shim thickness to meet the requirements of Drawing 639AS3169.

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5.1.1.6 MXM Axis intersection. Using Fixture SA 495242 and Deltronic tenth-pin (see 5.1.1.1), check for axis intersection. If common centerline of trunnions does not intersect spin axis, bore within requirements of the fixture adjust shim thickness between outer gimbal (Item 2, Figure 3) and inner gimbal (Item 1, Figure 3). When the common centerline of the trunnions are centered as required, and assembly has axial play as specified on the drawing, gimbal bearings are properly preloaded.

5.1.1.7 AI Starting torque. Using Fixture SA 495239, verify the maximum starting torque about the gimbal axis in accordance with Drawing 639AS3169.

5.1.2 AI Assembling outer axis assembly (see Figure 4). Proceed as follows:

5.1.2.1 MXM Outer gimbal. Assemble gimbal support ring (Item 1, Figure 4); two shims (Item 2, Figure 4), two gimbal screws (Item 5, Figure 4), and two gimbal bearings (Item 3, Figure 4), to inner axis assembly (Item 4, Figure 4). Select gimbal bearings in accordance with 5.1.1.4. Torque gimbal screws 16-18 inch-ounces. Do not cement screws.

5.1.2.2 MXM Shim thickness. Vary shim thickness as required to obtain proper intersection so that position of the nominal 0.480-inch counterbore of gimbal support ring relative to the nominal spin axis bore of inner gimbal, measured along geometric axis of 0.1875-inch bore in support ring (Item 1, Figure 4), is within the requirements of Drawing 639AS3169. Fixture SA 495242 may be used. Place assembly as shown on Fixture SA 2523897.

5.1.2.3 MXM Axial play. Measure axial play in outer pair of gimbal bearings (Item 2, Figure 4). Adjust axial play to meet the requirements of Drawing 639AS3169. Obtain play by adjusting shim thickness under outer pair of gimbal bearings (Item 3, Figure 4).

5.1.2.4 MXM Outer axis assembly. Remove outer axis assembly (Fixture SA 2523897). Place assembly as shown on Fixture SA 495242.

5.1.2.5 MXM Inner gimbal axis. Confirm center intersection (see 5.1.2.2). Maintain axial play requirements so that axial play of inner pair of gimbal bearings (Item 4, Figure 3) is still within limits specified. If inner gimbal axis has changed to a condition exceeding the specified requirements, adjust accordingly before continuing.

5.1.2.6 AI Torque. Verify that the maximum starting and rotational torque about each gimbal is as specified on Drawing 639AS3169.

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5.1.2.7 OP Install gimbal post. Temporarily install a gimbal post (Item 1, Figure 5) into outer axis assembly (Item 2, Figure 5). Use Fixture SA 493289 and Fixture SA 492478-3. Locking torque is 48-52 inch-ounces.

5.1.2.8 OP Deltronic gage pin. Insert an appropriately sized Deltronic gage pin into inner gimbal bore approximately 0.250 inch from the outer end of the inner gimbal. Place gimbal post flange surface against suitable mounting plate and project a 10X magnification image onto a 14-inch-minimum-diameter-screen shadowgraph. Observe angle of gimbal at limit by observing Deltronic gage pin shadow. Verify that the gimbal angle does not exceed 40 degrees at each 90-degree position; i.e., one gimbal axis in line with light beam and then rotated, in turn, 90 degrees at a time, until all four positions are checked. Verify that inspection sampling is in accordance with MIL-STD-105, AQL 4.0, Special Inspection Level S-3.

5.1.2.9 M Gimbal bearing bore. Check to verify that inner gimbal bearing bore has not been damaged by galling or other types of damage that will impair the usefulness of the gimbal assembly. Use a 20X or higher power microscope. Evidence of damage requires that the part either be repaired or scrapped.

5.1.2.10 MXM Ultrasonic cleaning. Clean assembly ultrasonically for 5 minutes.

5.1.2.11 AI Gimbal post. Remove gimbal post (Item 1, Figure 5) from assembly.

5.1.2.12 MXM Gimbal screws. Carefully remove the temporarily installed gimbal screws. (Item 5, Figure 3 and Item 5, Figure 4).

5.1.2.13 AI Thread cleaning. With solvent, carefully clean 1-72 female threads prior to cementing the screws (see Figure 6). Dry thoroughly.

5.1.2.14 AI Gimbal screw cleaning. Clean four gimbal screws with solvent. Dry thoroughly. To prevent stripping, maintain all socket wrenches in such a condition that a snug fit is ensured during installation.

5.1.2.15 MXM Adhesive application, thread holes. Apply adhesive to first two threads of each 1-72 thread hole (see Figure 6). Do not contaminate gimbal bearings with adhesive during this operation. Handle any loose fitting parts (i.e., high side of clearance) individually so they will not inadvertently be disassembled during cleaning and cementing.

5.1.2.16 MXM Adhesive application, gimbal screws. Apply adhesive to first two threads of each gimbal screw (see Figure 7).

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CAUTION MXM

Apply adhesive on first two threads only: more may cause contamination. As each gimbal screw seats with 16-18 inch-ounces of locking torque, the aft side of the hold will usually discharge a small amount of adhesive. Remove this excess with dry swabs, taking care not to contaminate gimbal bearings with lint from swabs or excess adhesive (see 4.5.4).

5.1.2.17 MXM Torque gimbal screws. Install the four gimbal screws. Torque all four screws 16-18 inch-ounces.

5.1.2.18 M Adhesive curing. Cure adhesive in accordance with manufacturers instructions. Accelerated curing temperature may not exceed 90°C (see 4.5.1).

5.1.2.19 M Bearing examination. On completion of cure, examine bearings under 20X microscope. Any form of contamination that cannot be removed is cause for rejection.

5.1.2.20 AI Drying. Dry assembly in oven at 60 ±5°C for 4 to 6 minutes.

5.1.2.20.1 OP Lubrication. Put excess lubricating oil (MIL-L-6085) into each gimbal bearing for protection during test of 5.1.2.21, using a syringe with filter holder and 0.45 - 1.2 micrometer filter for all gimbal bearing lubrication.

5.1.2.21 M Backoff torque. Apply counterclockwise backoff torque to each gimbal screw (Item 5, Figure 3 and Item 5, Figure 4) as required by Drawing 639AS3169. Any counterclockwise movement is cause for rejection.

NOTE: MXM Remove excess oil from gimbal bearings (using solvent) before final application of new, clean oil.

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5.1.2.22 MXM Gimbal bearing lubrication. Lubricate the four gimbal bearings as follows:

5.1.2.22.1 MXM Lubrication rotation. Place two drops of oil specified in MIL-L-6085 into each gimbal bearing using syringe with filter and No. 26 hypodermic needle. After application, rotate inner ring and balls.

5.1.2.22.2 MXM Removing excess oil. Remove excess oil from gimbal bearings by pulling a vacuum of 25 to 27 inches of mercury through each bearing for 5 to 15 seconds (see 4.5.2).

5.1.2.22.3 MXM Oil film removal. Remove oil film from gimbal body surfaces with solvent swab, taking care that gimbal bearings are not contaminated (see 4.5.4).

5.1.2.23 MXM Gimbal torque parameters. Confirm that gimbal torque about each axis is within requirements of Drawing 639AS3169. Reject or repair those units not meeting requirements.

5.1.3 AI Completing gimbal assembly (see Figure 2). Proceed as follows:

5.1.3.1 MXM Contamination examination. Examine the internal threads of the inner gimbal (Item 1, Figure 3) and bearing retainer (Item 2, Figure 2) for contamination under 20X microscope. If contaminated, clean with swab dampened in solvent. Remove any lint (see 4.5.4).

5.1.3.2 MXM Bearing fit clearance. Select matched spin bearing set (Item 1, Figure 2) that provides 0.000050 to 0.000100-inch clearance between bearing OD and bore of inner gimbal. Considering that mating diameters are not perfectly round, bearing-fit clearance refers to maximum OD and minimum ID differences. Gimbal identity is to include a record of spin bearing ID so that proper fit will be assured in future assembly operations to gimbal assembly.

5.1.3.2.1 AI Spin bearing preload. Spin bearing preload may be verified prior to assembly using SA described on Fixture SA 493245.

5.1.3.3 MXM Flange installation. Install flanged half of matched spin bearing in outer axis assembly (see 4.5.2).

5.1.3.4 MXM Bearing retainer. Start bearing retainer (Item 2, Figure 2) into threads of inner gimbal. The bearing retainer is not a permanent part of this assembly. Use only clean parts and equipment.

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5.1.3.5 MXM Bearing seating and torque. Using Fixture SA 495231, preliminarily seat retainer to ensure that bearing has seated, and then torque to 52-54 inch-ounces.

5.1.3.6 MXM Spin bearing examination. Examine gimbal assembly in the spin bearing area under 20X or higher power microscope (see Figure 8). Ascertain that no contamination has entered the spin bearing. If contaminated, remove lint with vacuum probe. If necessary, use slender, needlelike tweezers (properly decontaminated in clean solvent) for removal of lint.

5.1.3.7 MXM Spin bearing installation. Install plain half of matched spin bearing (Item 1, Figure 2) in outer axis assembly.

5.1.3.7.1 AI Shipping screw and nut. Install shipping screw (Item 4, Figure 2) and shipping nut (Item 5, Figure 2) to assembly. Torque screw 30-32 inch-ounces.

5.1.3.7.2 AI Support post torque. Install support post without adhesive and apply 48-52 inch-ounces torque.

5.1.3.8 MXM Gimbal smoothness. Move each gimbal axis through its total range of 80 degrees. Next, holding inner gimbal at extreme precession limit, rotate bearing mount through 360 degrees in contact with gimbal stop. If any binding, or uneven motion is detected, the gimbal is a reject unless repaired to correct faulty assembly.

5.1.3.9 MXM Gimbal inspection. Using Fixtures SA 2523897, SA 493213, and SA 493214, confirm that each assembled gimbal unit passes axial gimbal preload, axis intersection, and starting torque requirements (Drawing 639AS3169).

5.1.3.10 OP Shipping screw and nut. Install shipping screw (Item 4, Figure 2) and shipping nut (Item 5, Figure 2) to assembly (Drawing 639AS3169).

5.1.3.11 M Contamination inspection. Examine bearings, and bearing areas under 20X minimum magnification, and all remaining area under 10X minimum magnification for contamination in excess of the following acceptance criteria.

a. Nonmetallic contamination - no greater than 0.003 inch.

b. Metallic contamination - no greater than 0.001 inch.

5.1.3.11.1 M Storage. Store in a closed, dust-free container until the gimbal assembly is installed in the next higher assembly.

5.1.3.11.2 M Gimbal packaging. Gimbal packaging, when required, is performed in accordance with Drawing 639AS3169.

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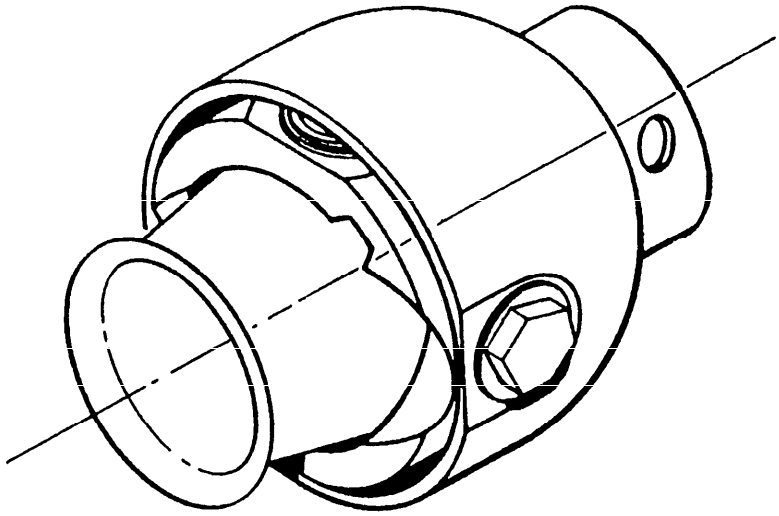
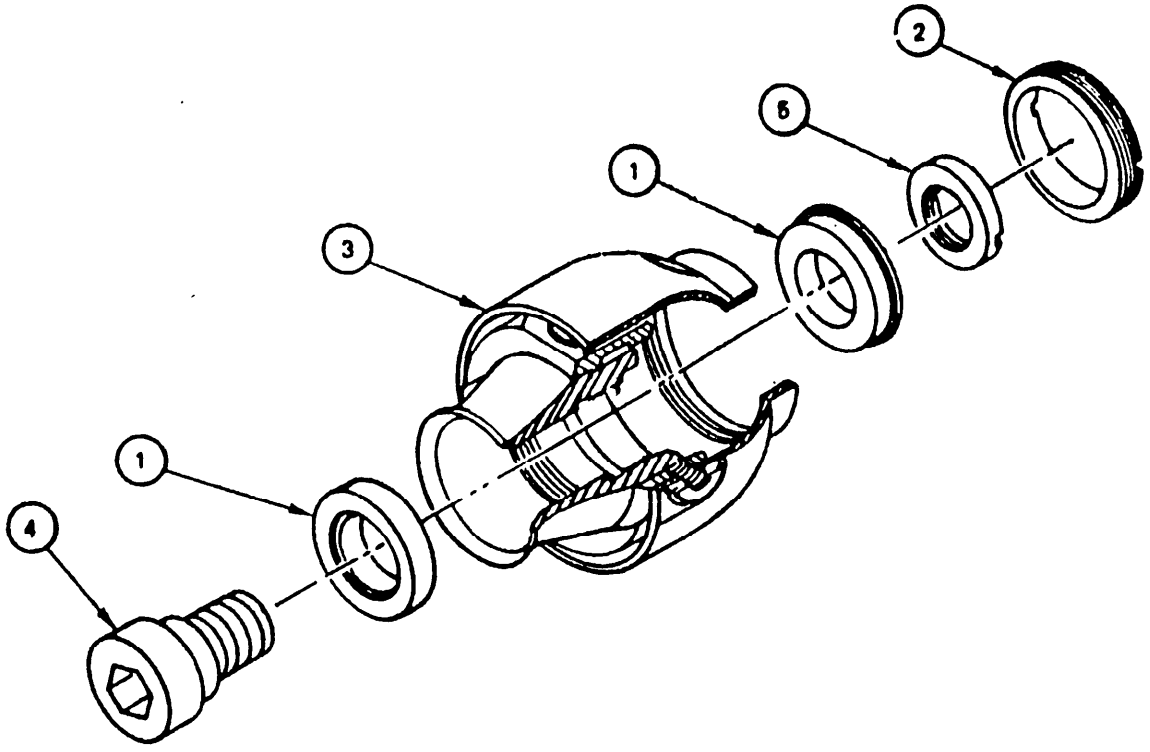


FIGURE 1. Gimbal assembly (Drawing 639AS3169).

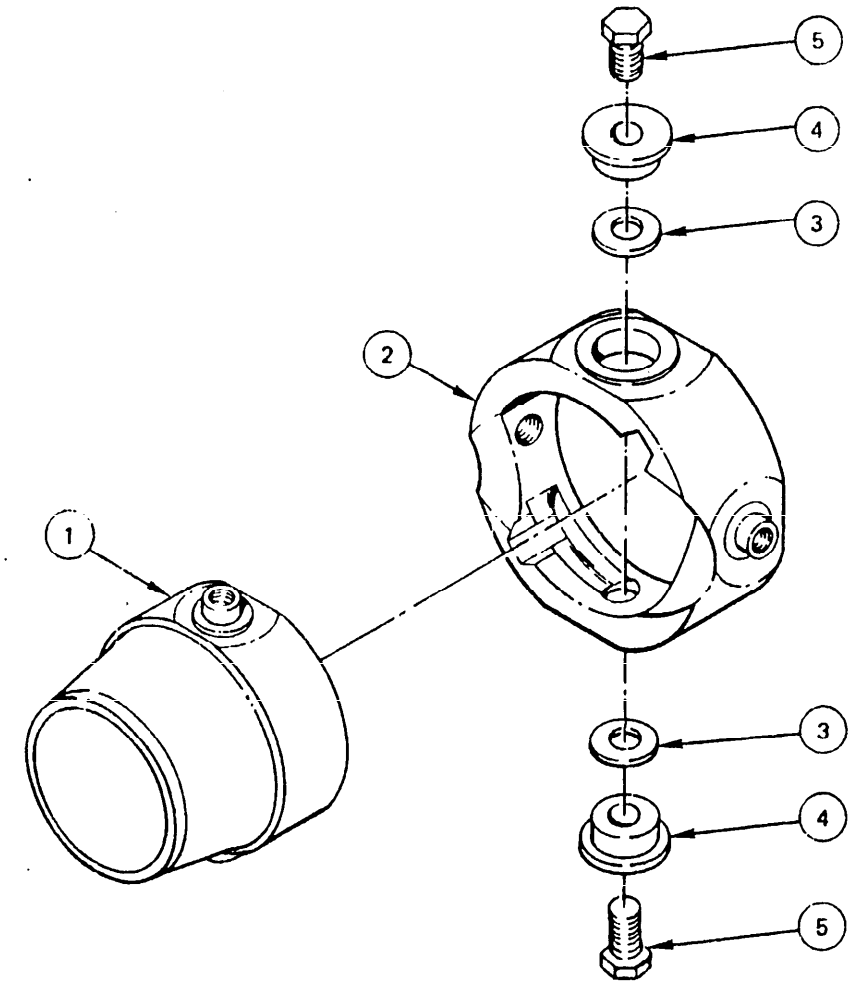
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ITEM	QTY	PART NUMBER	NOMENCLATURE
1	1	639AS3143	MATCHED SPIN BEARINGS
2	1	639AS1471	SHIPPING RETAINER
3	1	FIGURE 4	OUTER AXIS ASSEMBLY
4	1	639AS1469–1	SHIPPING SCREW
5	1	639AS1479–1	SHIPPING NUT

FIGURE 2. Exploded view of gimbal assembly (Drawing 639AS3169).

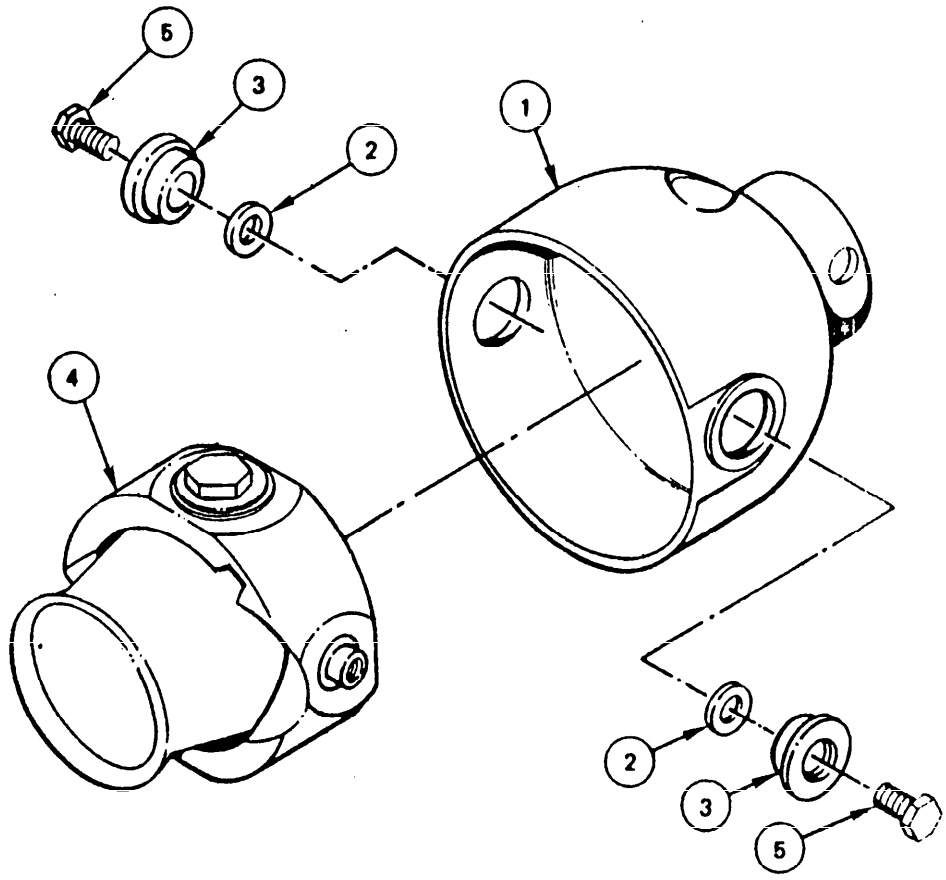
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ITEM	QTY	PART NUMBER	NOMENCLATURE
1	1	639AS3172	INNER GIMBAL
2	1	639AS624	OUTER GIMBAL
3	2AR	639AS627	GIMBAL BEARING SHIM
4	2	639AS3168	GIMBAL BEARING
5	2	639AS750	GIMBAL SCREW

FIGURE 3. Exploded view of inner axis assembly (Drawing 639AS3169).

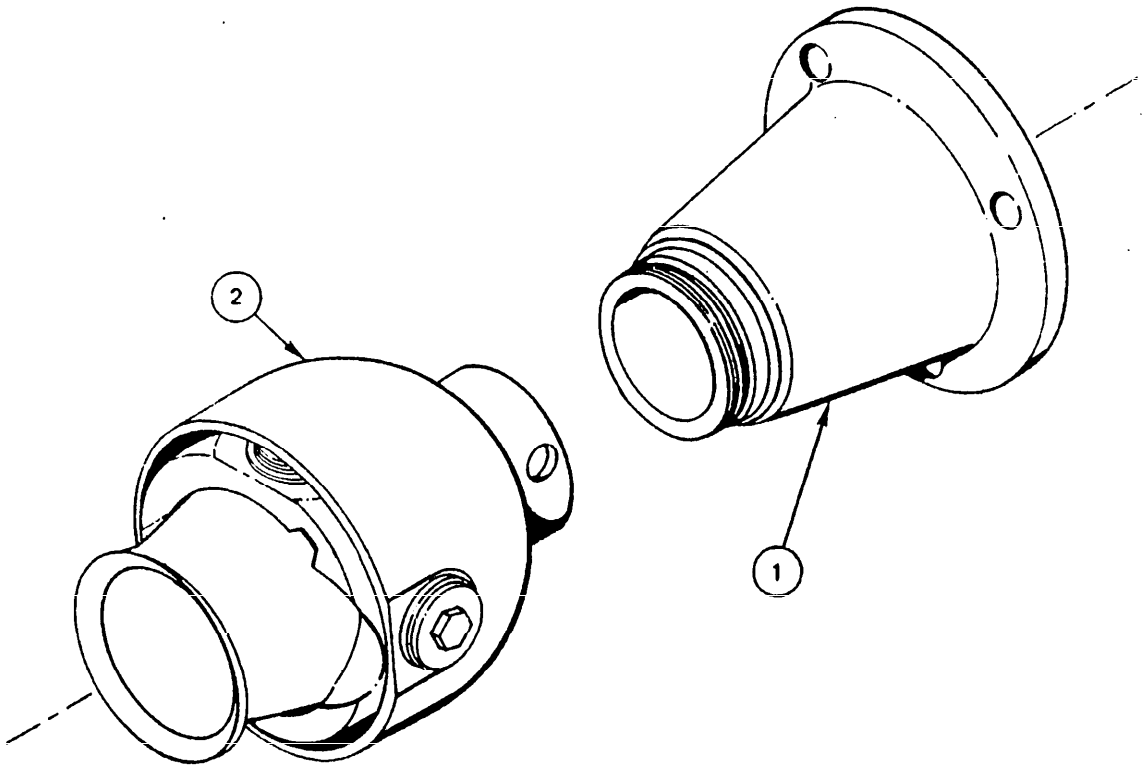
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ITEM	QTY	PART NUMBER	NOMENCLATURE
1	1	639AS551	RING, GIMBAL
2	2	639AS627	GIMBAL BEARING SHIM
3	1	639AS3168	GIMBAL BEARING
4	1	FIGURE 3	INNER AXIS ASSEMBLY
5	2	639AS750	GIMBAL SCREW

FIGURE 4. Exploded view of outer axis assembly (Drawing 639AS3169).

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ITEM	QTY	PART NUMBER	NOMENCLATURE
1	1	639AS537	GIMBAL POST
2	1	FIGURE 4	OUTER AXIS ASSEMBLY

FIGURE 5. Outer axis assembly and support post (Drawing 639AS3169).

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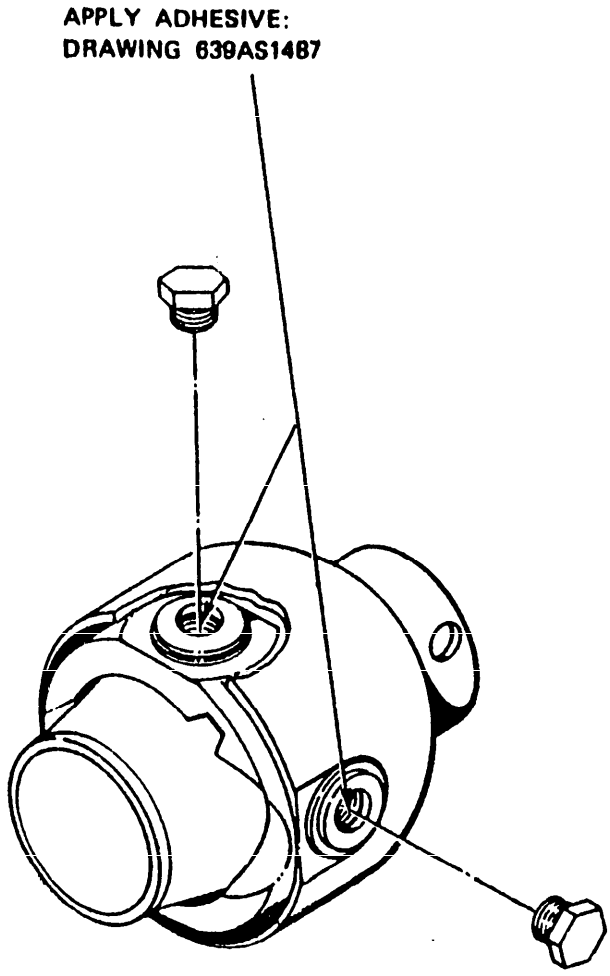


FIGURE 6. Applying adhesive to gimbals (Drawing 639AS3169).

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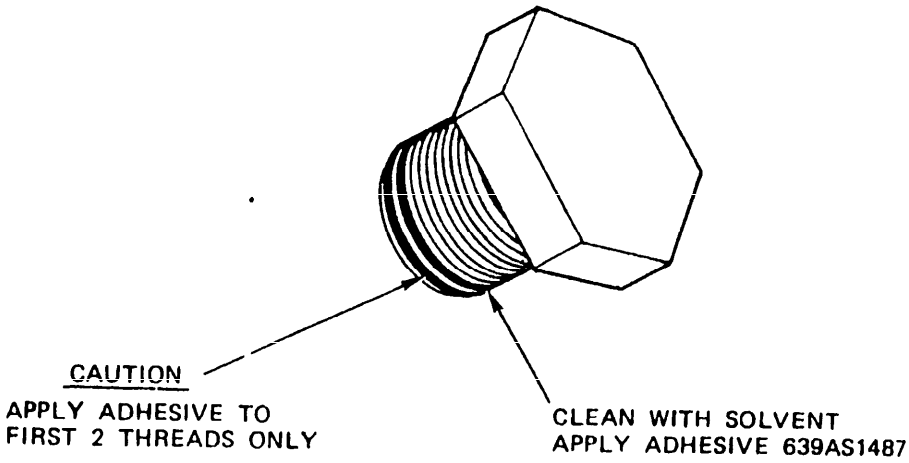


FIGURE 7. Applying adhesive to gimbal screws.

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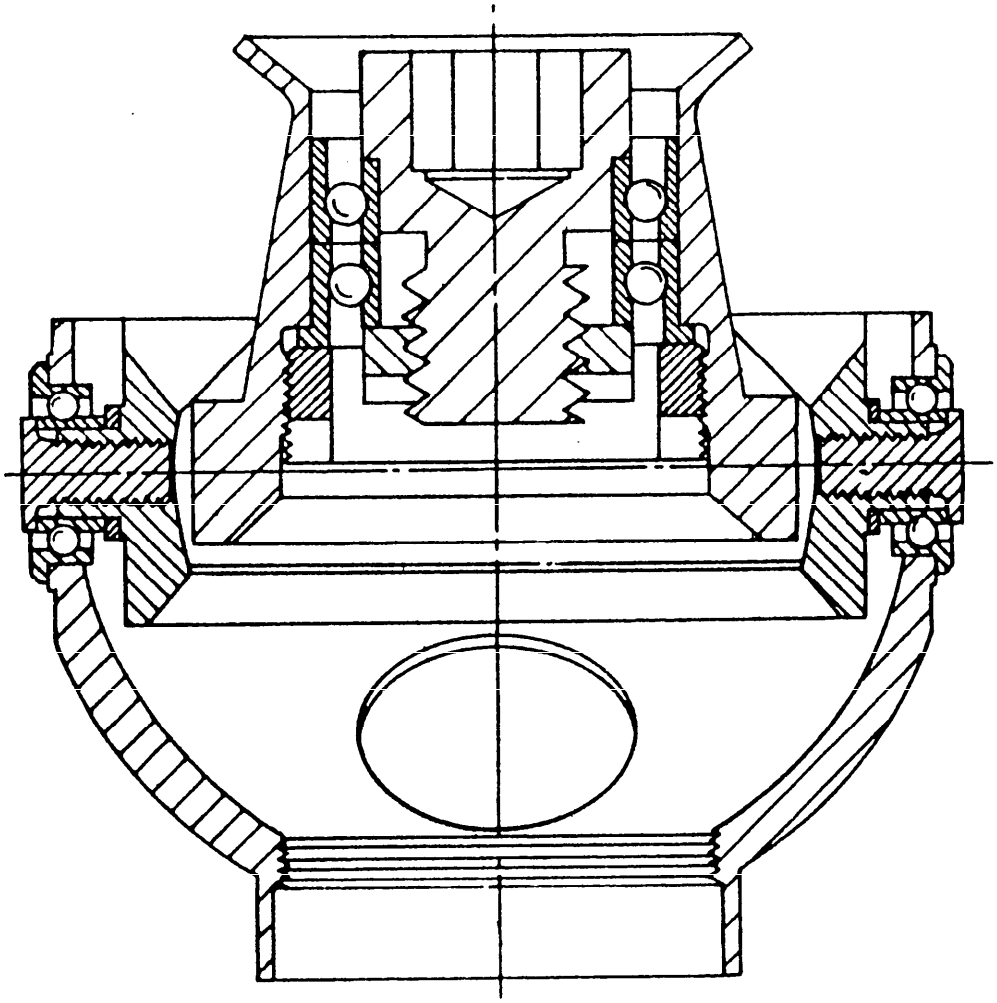


FIGURE 8. Gimbal assembly cross section (Drawing 639AS3169).

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